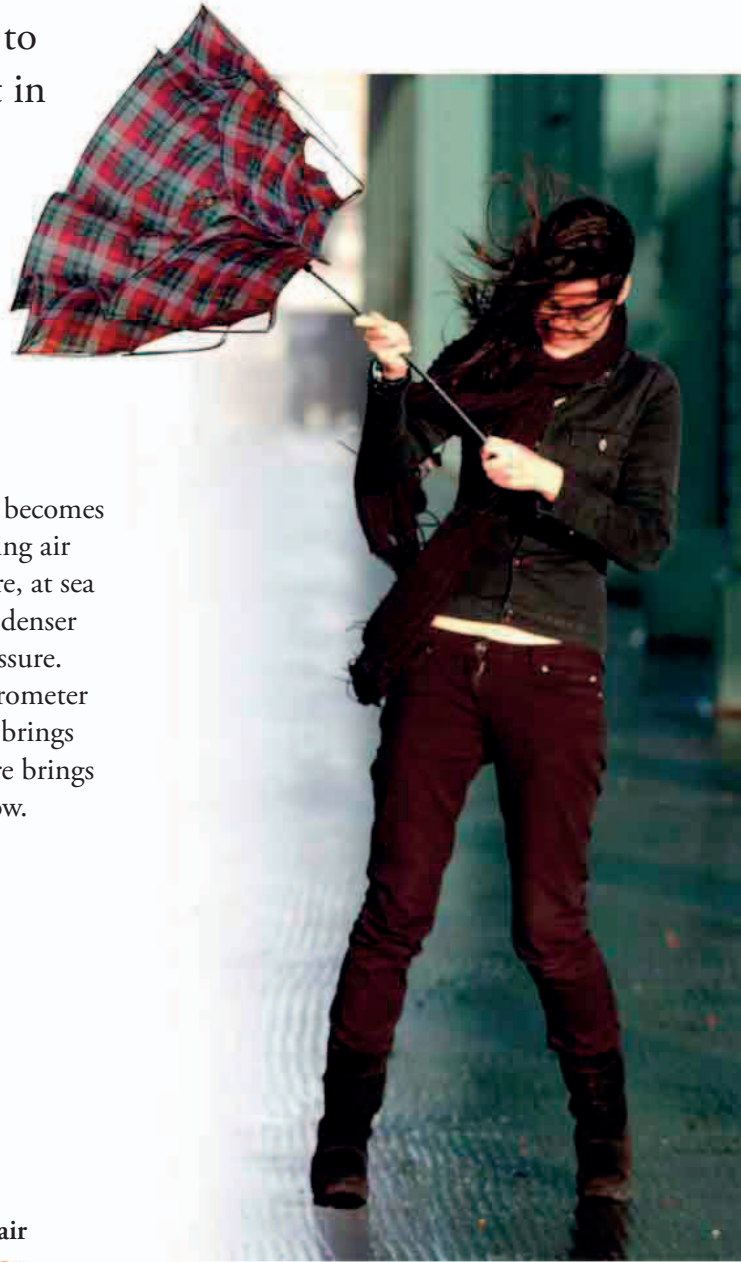


Weather systems

The weather can change dramatically from day to day, and even hour to hour. It changes very fast in cool oceanic regions such as northern Europe, where weather systems are carried in from the ocean. These bring rapid sequences of clouds, rain, storm winds, sunshine, and showers, making weather forecasting very difficult.

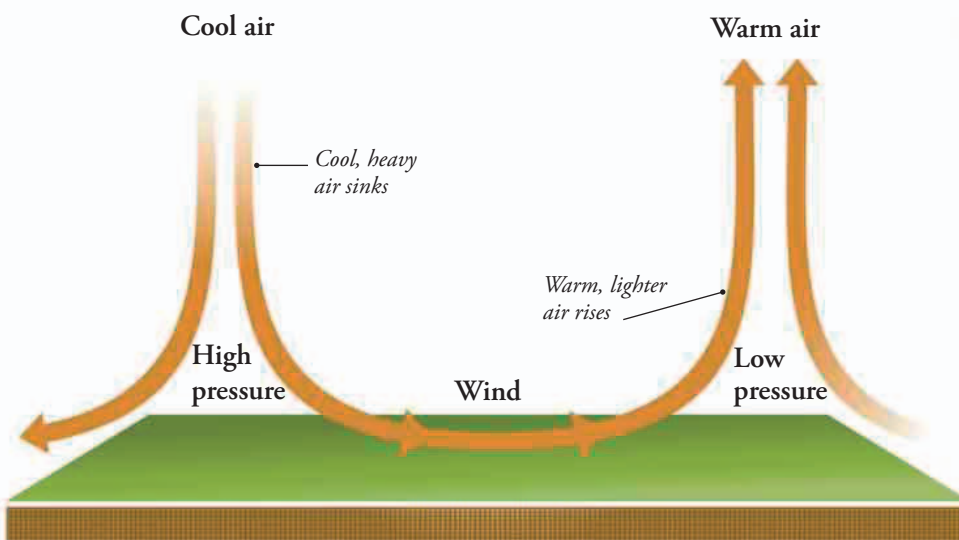


HIGHS AND LOWS

As air warms up, it expands, becomes less dense, and rises. This rising air reduces its weight, or pressure, at sea level. By contrast, cool air is denser and sinks, exerting more pressure. Pressure is measured by a barometer like this one—high pressure brings fine weather, but low pressure brings clouds, rain, storms, and snow.

FROM HIGH TO LOW

The atmospheric pressure in a zone of sinking cold air is higher than in a zone of rising warm air. This difference in pressure makes low-level air flow out of high-pressure zones into nearby low-pressure zones. The air is drawn upward as it reaches the center of the low-pressure zone, allowing more air to flow in behind it. We experience this flow of air as wind.



WIND STRENGTH

The bigger the pressure difference between neighboring highs and lows, the stronger the wind that blows between them. The strength of the wind is described using the Beaufort scale of wind force, which rises from 0 for a still day to 12 for hurricane force. It becomes difficult to use an umbrella around scale number 6.